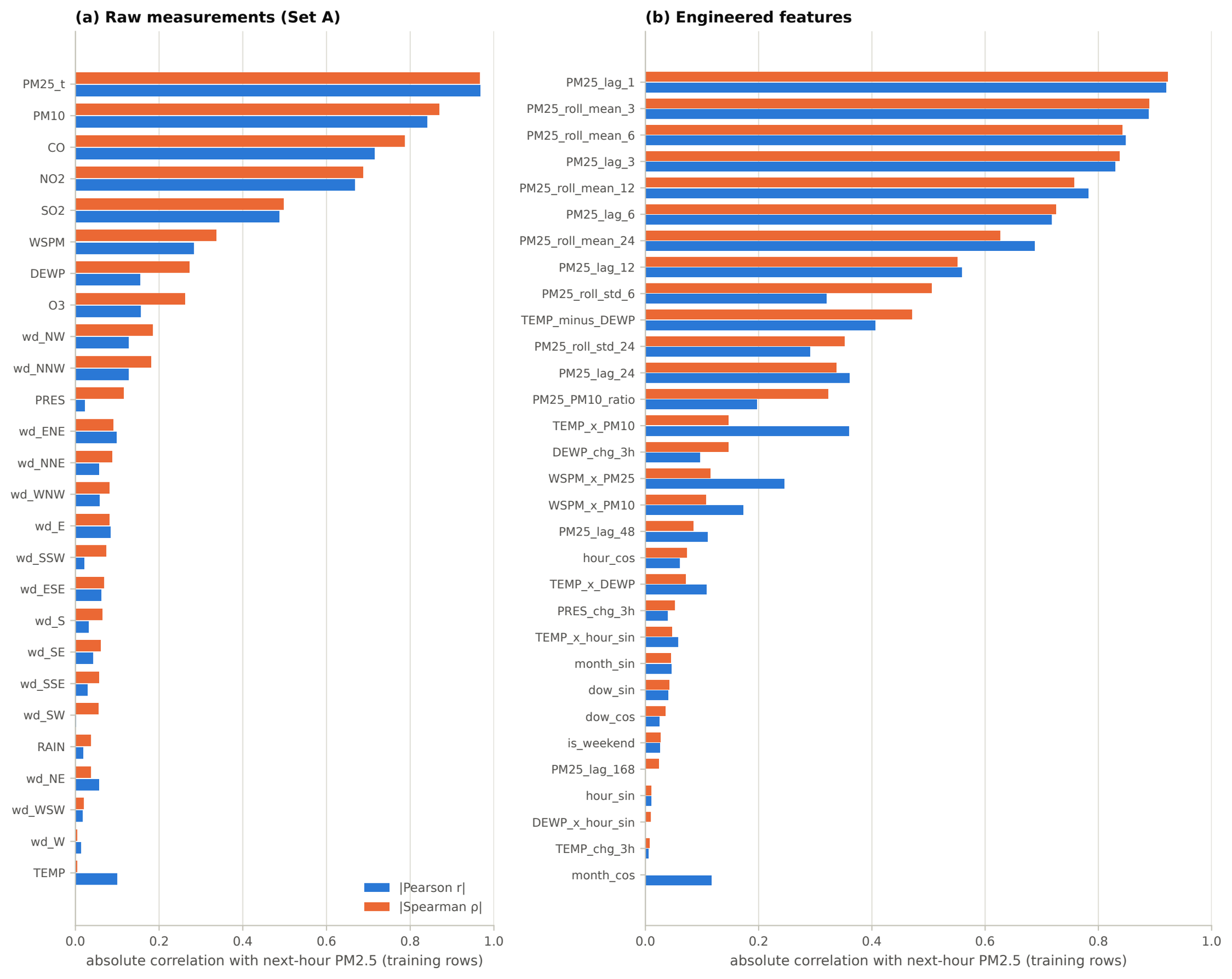
(a) Raw measurements (Set A)
PM25_t
PM10
CO
NO2
SO2
WSPM
DEWP
O3
wd_NW
wd_NNW
PRES
wd_ENE
wd_NNE
wd_WNW
wd_E
wd_SSW
wd_ESE
wd_S
wd_SE
wd_SSE
wd_SW
RAIN
wd_NE
wd_WSW
wd_W
TEMP
|Pearson r|
|Spearman ρ|
0.0
0.2
0.4
0.6
0.8
1.0
absolute correlation with next-hour PM2.5 (training rows)
(b) Engineered features
PM25_lag_1
PM25_roll_mean_3
PM25_roll_mean_6
PM25_lag_3
PM25_roll_mean_12
PM25_lag_6
PM25_roll_mean_24
PM25_lag_12
PM25_roll_std_6
TEMP_minus_DEWP
PM25_roll_std_24
PM25_lag_24
PM25_PM10_ratio
TEMP_x_PM10
DEWP_chg_3h
WSPM_x_PM25
WSPM_x_PM10
PM25_lag_48
hour_cos
TEMP_x_DEWP
PRES_chg_3h
TEMP_x_hour_sin
month_sin
dow_sin
dow_cos
is_weekend
PM25_lag_168
hour_sin
DEWP_x_hour_sin
TEMP_chg_3h
month_cos
0.0
0.2
0.4
0.6
0.8
1.0
absolute correlation with next-hour PM2.5 (training rows)